

Natural Boundaries of Quadratic Exponential Feedback

An exact obstruction to angular Borel summation,
analytic-curve rigidity, and a rational-amplitude dichotomy

Research article prepared for Vladimir Reshetnikov
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Abstract

We study the formal feedback equation

$$\widehat{U}(q) = \sum_{j \geq 1} q^j e^{j^2 \widehat{U}(q)}$$

and the literal left-side realization of its compositional inverse, characterized by $\sum_{j \geq 1} Q(u)^j e^{j^2 u} = u$. We prove that a neighborhood of zero in the imaginary axis is a natural boundary for Q , although Q is smooth up to that boundary. Its integrated Borel transform is entire and exponentially bounded in every fixed-width neighborhood of the negative ray, but has no exponential bound in any open angular neighborhood of that ray. Consequently $\widehat{Q}$, and also $\widehat{U}$, is not angularly 1-summable in direction π . This answers negatively the specific open-arc question in the repository's negative-ray summation article.

The main tool is a local rigidity theorem for a heat expansion restricted to an arbitrary analytic curve. If the initial meromorphic function has nearest poles at distances s_ν , then the coefficient contribution of a pole of order p retains the nonzero multiplier $\exp(-a'(0)s_\nu/2)$ under the curve $a = a(t)$. Distinct squared pole distances prevent persistent cancellation. At rational imaginary times the quadratic kernel has rational generating data, so an analytic implicit inverse would contradict this rigidity theorem.

We extend the argument to rational amplitude generating functions and eventually quadratic real slopes. Within this class, angular negative-direction summability holds exactly for finite action support. Nonpolynomial rational amplitudes instead give a smooth natural boundary. Finite-action inverses nevertheless converge exponentially, with every fixed number of derivatives, on a closed left half-disc. Full proofs, exact-arithmetic checks, 120-digit numerical diagnostics, and further research questions are included.

Status and scope. These are conventional mathematical proofs, not Lean-verified or externally refereed results. The proposed contribution is the implicit natural-boundary theorem and its consequences for the stated feedback class. Classical heat summability, partial-theta methods, and factorial-composition mechanisms are credited rather than claimed as new. Global publication priority is not established. No general impossibility theorem for transseries summation is asserted.

Contents

1	The repository question and the distinction it requires	3
2	Main statements and conventions	4
3	The literal inverse on a closed half-disc	5
4	Factorial expansions along analytic curves	7
5	A heat-pole rigidity theorem	9
6	Rational phases force a natural boundary	10
7	An entire Borel transform and its fine sum	11
8	Why no angular neighborhood can work	14
9	Rational amplitudes and robustness	15
10	Finite-action approximation does not detect the boundary	17
11	Examples, checks, and limitations	18
12	Further research questions	19
13	Conclusion	21
A	Proof-dependency and claim audit	21
B	Reproducibility and source provenance	22

1 The repository question and the distinction it requires

1.1 The exact question answered

The ProveIt repository contains a transseries development and a collection of research articles on countable exponential feedback. The snapshot inspected for this article is

0c973d8f5e7ef4550f4df1bf486d890037fd9f14.

The immediate source is *Negative-Ray Summation of Countable Exponential-Feedback Transseries* [2]. Its research agenda, in the subsection on angular summability of the quadratic model, asks whether an open arc of Borel directions containing π can have the uniform exponential-growth bounds required for angular summation. The source explicitly distinguishes that problem from its proved fixed-width-strip result. The answer obtained here is *no*.

The distinction is essential. An entire Borel transform has no finite Borel singularities, but can still grow too rapidly away from one ray to permit rotation of the Laplace contour through any nonzero angle. Conversely, an analytic function in a half-disc can be infinitely differentiable on its diameter without admitting analytic continuation through even one point of that diameter. Our example realizes both phenomena simultaneously, in a nonlinear implicit equation.

The related regularity and negative-ray articles already establish formal existence, coefficient-growth information, and fine summation [3, 2]. We do not claim those as new. For the quadratic inverse we rederive the needed Borel representation and its fine-summation property, with particularly simple frequency bounds. This makes the main negative result independent of unverified asymptotic claims in other research drafts.

1.2 What is not answered

We do not determine the set of individual rays on which the Borel transform has finite exponential type. An isolated ray is not an open arc. We do not compute its precise growth in shrinking angular neighborhoods, the best Taylor-truncation error, or nonlinear Stokes constants. We also do not infer differential transcendence from a natural boundary: nonlinear differential equations can have solutions with natural boundaries.

Nor does the result contradict resurgence understood solely as suitable analytic continuation of a Borel germ. An entire Borel transform already has unrestricted finite-plane continuation. The obstruction here concerns growth at infinity and the resulting sectorial Laplace sums, not the existence of isolated finite Borel singularities.

1.3 Relation to classical work

The formal heat equation with meromorphic initial data is a classical source of Gevrey expansions and Stokes phenomena. Lutz, Miyake, and Schäfer established foundational summability results for divergent heat solutions [5]; Michalik and Podhajecka analyze Stokes phenomena for meromorphic heat data [6]. Han, Li, Sauzin, and Sun study partial theta series, rational boundary points, discrete Fourier transforms, and resurgence [7]. Sauzin distinguishes fine summation from summation in an arc and proves nonlinear stability results [4].

These works explain why heat poles and rational phases are natural tools. The extra issue here is *implicit cancellation*: the quantity playing the role of the exponential amplitude is itself an unknown function of time. A fixed-amplitude gap theorem alone does not exclude the possibility that this function removes a singularity. Sections 4 and 5 prove the analytic-curve obstruction needed to exclude precisely that possibility.

2 Main statements and conventions

We use $\widehat{f}$ for a formal power series and f for an analytic or boundary-smooth realization. The integrated Borel transform is

$$\mathcal{B}_0\left(\sum_{n \geq 0} f_n u^n\right)(\zeta) = \sum_{n \geq 0} \frac{f_n}{n!} \zeta^n. \quad (2.1)$$

Differentiating this transform gives the usual Borel minor of a series with zero constant term.

Definition 2.1 (Fine and angular negative-direction summability). For $\delta > 0$ write

$$T_\delta = \{\zeta : \text{dist}(\zeta, (-\infty, 0]) < \delta\}.$$

A series is *finely summable in direction π* if its Borel germ continues to some T_δ and there obeys $|\mathcal{B}_0 \widehat{f}(\zeta)| \leq C e^{H|\zeta|}$. It is *angularly 1-summable in direction π* if such a bound holds in an open sector $|\arg \zeta - \pi| < \eta$ for some $\eta > 0$, together with a neighborhood of the origin. The argument is understood on a continuous determination around π . Equivalent definitions allow locally bounded exponential constants as the direction varies; restricting to a smaller closed arc recovers the uniform formulation used here.

On the negative input ray the fine sum is

$$\mathcal{S}_\pi \widehat{f}(-r) = \frac{1}{r} \int_0^\infty e^{-s/r} \mathcal{B}_0 \widehat{f}(-s) ds, \quad 0 < r \ll 1. \quad (2.2)$$

A fixed-width Borel neighborhood gives an input domain tangent to the origin. An angular Borel neighborhood permits rotated Laplace contours and produces an input sector of opening strictly greater than π . We give the precise contour argument used here in Section 8.

Definition 2.2 (Natural boundary along an arc). Let f be holomorphic on one side of an open boundary arc I . We say that I is a natural boundary for this branch of f if no point of I has a complex neighborhood on which a holomorphic function agrees with f on the original side. This is a local statement about the specified branch, not a claim about an unrelated function having the same formal asymptotic expansion.

Put

$$K(q, u) = \sum_{j \geq 1} q^j e^{j^2 u}, \quad r_0 = \frac{1}{32}, \quad D_- = \{u : |u| < r_0, \text{Re } u < 0\}. \quad (2.3)$$

We use boundary values on $\text{Re } u = 0$ whenever $|u| < r_0$.

Theorem 2.3 (Quadratic feedback: smooth boundary, no angular sum). *There is a unique small solution Q of*

$$K(Q(u), u) = u, \quad |Q(u)| \leq \frac{1}{16}, \quad |u| \leq \frac{1}{32}, \quad \text{Re } u \leq 0. \quad (2.4)$$

It is holomorphic in D_- , has continuous derivatives of every order on the closed half-disc, and satisfies $Q(0) = 0$, $Q'(0) = 1$, and

$$|Q'(u) - 1| < \frac{1}{3}. \quad (2.5)$$

The following conclusions hold.

- (i) *Every point of $\{it : |t| < r_0\}$ is a natural-boundary point for Q .*

- (ii) *The formal Taylor series $\widehat{Q}$ is the compositional inverse of the unique formal solution $\widehat{U}$ of $\widehat{U}(q) = K(q, \widehat{U}(q))$. Its integrated Borel transform is entire and exponentially bounded on every fixed-width T_δ . Its fine sum is Q .*
- (iii) *For every $\eta > 0$ and every $H > 0$,*

$$\sup_{\substack{\zeta \neq 0 \\ |\arg \zeta - \pi| < \eta}} e^{-H|\zeta|} |\mathcal{B}_0 \widehat{Q}(\zeta)| = \infty. \quad (2.6)$$

Neither $\widehat{Q}$ nor $\widehat{U}$ is angularly 1-summable in direction π .

- (iv) *The inverse function $U = Q^{(-1)}$ has the smooth arc $\Gamma = \{Q(it) : |t| < r_0\}$ as a natural boundary. This arc is nowhere real-analytic as a regular plane curve.*

The words “every fixed-width” do not imply a common exponential constant: the constant can grow with δ . The impossibility in (2.6) concerns wedges of fixed positive angle, whose width grows proportionally to their distance from zero.

The next theorem is a classification, not merely a perturbative example.

Theorem 2.4 (Rational-amplitude dichotomy). *Let*

$$\mathcal{A}(z) = \sum_{j \geq 1} c_j z^j \quad (2.7)$$

be rational and holomorphic at zero, with $c_1 \neq 0$. Suppose the real slopes satisfy, for all sufficiently large j ,

$$\lambda_j = \alpha j^2 + \beta j + \gamma, \quad \alpha > 0, \quad (2.8)$$

with arbitrary finite real modifications at smaller indices. The small left-side inverse defined by

$$\sum_{j \geq 1} c_j Q(u)^j e^{\lambda_j u} = u \quad (2.9)$$

exists on a sufficiently small left half-disc and is smooth up to its diameter. Its formal series has an entire integrated Borel transform and is finely summable in direction π , with sum Q .

The following conditions are equivalent:

- (a) *$\mathcal{A}$ is a polynomial, equivalently the action amplitudes have finite support;*
- (b) *Q extends holomorphically to a neighborhood of zero;*
- (c) *the formal series $\widehat{Q}$ has positive radius of convergence;*
- (d) *$\widehat{Q}$ is angularly 1-summable in direction π .*

If these conditions fail, every point of a sufficiently small imaginary interval is a natural-boundary point for Q .

The rationality hypothesis is substantive. Rapidly decaying nonrational amplitudes can make the kernel holomorphic across the boundary; an example appears in Section 11. Thus the theorem does not replace a joint amplitude–slope analysis for arbitrary amplitudes.

3 The literal inverse on a closed half-disc

3.1 Uniform contraction with explicit constants

For $\operatorname{Re} u \leq 0$ write (2.4) as

$$q = T_u(q) := e^{-u} \left(u - \sum_{j \geq 2} q^j e^{j^2 u} \right). \quad (3.1)$$

When $|q| \leq 1/16$ and $|u| \leq 1/32$,

$$|T_u(q)| \leq \frac{32}{31} \left(\frac{1}{32} + \frac{1}{240} \right) = \frac{17}{465} < \frac{1}{16}.$$

Here $e^{1/32} \leq 32/31$ follows from $e^x \leq (1-x)^{-1}$ for $0 \leq x < 1$. Moreover,

$$|\partial_q T_u(q)| \leq \frac{32}{31} \left(\frac{1}{(1-1/16)^2} - 1 \right) = \frac{32}{225} < 1. \quad (3.2)$$

The contraction theorem gives the unique solution uniformly on the closed parameter half-disc. Iterating from zero gives holomorphic functions on the open half-disc; uniform convergence proves holomorphy of their limit. Continuity up to the diameter follows from the same uniform convergence.

3.2 Smooth boundary jets and injectivity

Every mixed derivative of K is represented, on a slightly smaller q -disc, by a uniformly convergent series of the form

$$\sum_{j \geq 1} (j)_s j^{2t} q^{j-s} e^{j^2 u}, \quad \operatorname{Re} u \leq 0.$$

Polynomial factors in j are absorbed by the geometric decay in q . The denominator K_q stays bounded away from zero. Implicit differentiation, first in the interior and then by continuous extension, therefore gives all boundary derivatives of Q . Tangential derivatives also follow directly by differentiating the uniformly contracting boundary equation. Repeating this argument proves that the one-sided complex derivatives and the boundary tangential derivatives agree. This is the precise meaning of smoothness on the closed half-disc.

For the quantitative first derivative, implicit differentiation gives

$$Q'(u) = \frac{1 - K_u(Q(u), u)}{K_q(Q(u), u)}. \quad (3.3)$$

Set

$$d = \frac{1}{31} + \frac{31}{225}, \quad e = \frac{272}{3375}.$$

The elementary geometric sums yield

$$\begin{aligned} |K_q - 1| &\leq |e^u - 1| + \sum_{j \geq 2} j(1/16)^{j-1} \leq d, \\ |K_u| &\leq \sum_{j \geq 1} j^2(1/16)^j = \frac{(1/16)(1 + 1/16)}{(1 - 1/16)^3} = e. \end{aligned}$$

Since $(d + e)/(1 - d) < 1/3$, equation (3.3) proves (2.5). Integrating $Q' - 1$ on line segments in the convex half-disc gives

$$\frac{2}{3}|u - v| \leq |Q(u) - Q(v)| \leq \frac{4}{3}|u - v|. \quad (3.4)$$

The same inequality holds on the closure by continuity. In particular, Q is univalent, its boundary parametrization is regular, and $Q(u) = 0$ only when $u = 0$. The last fact also follows immediately from $K(0, u) = 0$.

3.3 Formal identification

The operator

$$V \mapsto \sum_{j \geq 1} q^j e^{j^2 V}$$

improves agreement of series in $q\mathbb{C}[[q]]$ by one order. It therefore has a unique formal fixed point $\hat{U}$ with linear coefficient one. Its formal compositional inverse exists. Substituting that inverse into the fixed point equation gives the formal version of (2.4); conversely that equation determines its coefficients successively. The boundary derivatives just constructed satisfy the same finite coefficient equations, so their Taylor series is $\hat{Q} = \hat{U}^{(-1)}$.

The beginning is

$$\hat{Q}(u) = u - 2u^2 + \frac{1}{2}u^3 - \frac{2}{3}u^4 - \frac{29}{8}u^5 - \frac{743}{60}u^6 - \frac{8491}{144}u^7 - \frac{92726}{315}u^8 + \dots \quad (3.5)$$

This identity is formal; it does not assert convergence.

4 Factorial expansions along analytic curves

We now prove the estimate that prevents an unknown analytic amplitude from hiding a heat singularity. The proof is elementary and includes the uniform bound needed to justify the limiting coefficient sum.

For an integer $p \geq 1$ define

$$D_{p,n} = \frac{\Gamma(2n+p)}{\Gamma(p)n!}, \quad F_p(z) = \sum_{n \geq 0} D_{p,n} z^n. \quad (4.1)$$

The second expression is a formal series. Differentiating a pole of order p twice at each heat step produces exactly these coefficients.

Lemma 4.1 (Analytic composition of a factorial series). *Let $p \geq 1$ be an integer, let $s(t)$ be analytic near zero with $s(0) = s_0 \neq 0$, and put $s_1 = s'(0)$. Then*

$$[t^n] \left\{ \sum_{\ell \geq 0} \frac{D_{p,\ell} t^\ell}{s(t)^{2\ell+p}} \right\} = D_{p,n} s_0^{-2n-p} \left(e^{-s_0 s_1/2} + o(1) \right). \quad (4.2)$$

The $o(1)$ is additive inside the displayed parentheses. In particular, it does not assert a relative error for a sum of several pole contributions that might be small by oscillation.

Proof. Let $R(t) = s_0^2/s(t)^2$ and $h(t) = s(t)^{-p}$. Thus $R(0) = 1$ and $R'(0) = -2s_1/s_0$. If b_n is the coefficient on the left of (4.2), then the finite coefficient identity

$$\frac{b_n}{D_{p,n} s_0^{-2n}} = \sum_{k=0}^n \frac{D_{p,n-k}}{D_{p,n}} s_0^{2k} [t^k] \{h(t) R(t)^{n-k}\} \quad (4.3)$$

follows by setting $k = n - \ell$.

For fixed k , the exact ratio

$$\frac{D_{p,m+1}}{D_{p,m}} = \frac{(2m+p)(2m+p+1)}{m+1}$$

gives $D_{p,n-k}/D_{p,n} \sim (4n)^{-k}$. Since $R(0) = 1$, the coefficient $[t^k]\{h(t)R(t)^{n-k}\}$ is a polynomial in n of degree at most k , with leading coefficient $h(0)R'(0)^k/k!$. The k th summand in (4.3) therefore tends to

$$h(0) \frac{(s_0^2 R'(0)/4)^k}{k!}.$$

To justify summation of these limits, first note that, for all $m \geq 0$ and $p \geq 1$,

$$\frac{D_{p,m+1}}{D_{p,m}} \geq 2(m+1).$$

The geometric mean of the largest k integers in $\{1, \dots, n\}$ is at least the geometric mean of all n of them, and $n! \geq (n/e)^n$. Consequently, for $1 \leq k \leq n$,

$$\frac{D_{p,n-k}}{D_{p,n}} \leq 2^{-k} \frac{(n-k)!}{n!} \leq \left(\frac{e}{2n}\right)^k. \quad (4.4)$$

Choose $\delta > 0$ small enough that R has an analytic logarithm on $|t| \leq \delta$. There are constants H, M such that $|h(t)| \leq H$ and $|R(t)| \leq e^{M|t|}$ there. Choose $B \geq \max(M, 1/\delta, 1)$. Cauchy's coefficient estimate on the circle $|t| = k/(Bn)$ gives

$$|[t^k]\{h(t)R(t)^{n-k}\}| \leq H \left(\frac{eBn}{k}\right)^k.$$

Combined with (4.4), the absolute value of the k th summand of (4.3) is at most

$$H \left(\frac{|s_0|^2 e^2 B}{2k}\right)^k. \quad (4.5)$$

This majorant is summable over $k \geq 1$ and independent of n . The term $k = 0$ is bounded separately. Dominated convergence in the counting measure now proves that (4.3) tends to

$$h(0) \exp\left(\frac{s_0^2 R'(0)}{4}\right) = s_0^{-p} e^{-s_0 s_1/2}.$$

This is (4.2). □

Remark 4.2 (Why the first derivative, but not higher derivatives, survives). For a fixed loss of k heat indices, the factorial ratio contributes n^{-k} and the coefficient of the analytic substitution contributes n^k . Only repeated uses of the linear coefficient of $R(t) - 1$ reach that highest degree in n . Higher derivatives of $s(t)$ affect lower orders. The exponential multiplier is never zero, which is the feature needed for a noncancellation theorem.

Lemma 4.3 (Analytic remainder estimate). *Let ψ be analytic on $|a - a_0| < R_*$ and let $a(t)$ be analytic near zero with $a(0) = a_0$. For every $R_1 < R_*$ there is a constant C such that, for all sufficiently large n ,*

$$\left| [t^n] \sum_{\ell \geq 0} \frac{t^\ell}{\ell!} \psi^{(2\ell)}(a(t)) \right| \leq C D_{1,n} R_1^{-2n}. \quad (4.6)$$

Proof. Choose R_2 with $R_1 < R_2 < R_*$. If $\psi(a_0 + x) = \sum c_m x^m$, Cauchy's estimate gives $|c_m| \leq M R_2^{-m}$. Write $a(t) - a_0 = \sum_{j \geq 1} a_j t^j$ and $A(t) = \sum_{j \geq 1} |a_j| t^j$. Coefficientwise absolute values of the series in (4.6) are bounded by the coefficients of

$$M R_2 \sum_{\ell \geq 0} \frac{D_{1,\ell} t^\ell}{(R_2 - A(t))^{2\ell+1}}.$$

Indeed $M R_2 / (R_2 - x)$ majorizes $\psi(a_0 + x)$, and taking an even number of derivatives and substituting A preserve this majorization. Lemma 4.1 bounds these coefficients by $C' D_{1,n} R_2^{-2n}$ for large n . This is stronger than (4.6). □

5 A heat-pole rigidity theorem

Let ϕ be meromorphic near a closed disc centered at a_0 , analytic at a_0 , and with at least one pole. Let $R > 0$ be the distance from a_0 to the nearest poles, and denote those poles by $\omega_1, \dots, \omega_d$. Put

$$s_\nu = a_0 - \omega_\nu, \quad |s_\nu| = R. \quad (5.1)$$

Suppose ϕ has no other singularity in some larger disc $|a - a_0| < R_*$, $R_* > R$. Let p be the largest pole order among the nearest poles, and let J index those of order p . For $\nu \in J$ write the highest Laurent term as $r_\nu(a - \omega_\nu)^{-p}$, with $r_\nu \neq 0$.

Theorem 5.1 (Rigidity under analytic curves). *For every analytic $a(t)$ with $a(0) = a_0$, the formal series*

$$\widehat{G}(t) = \sum_{\ell \geq 0} \frac{t^\ell}{\ell!} \phi^{(2\ell)}(a(t)) = \sum_{n \geq 0} g_n t^n \quad (5.2)$$

satisfies

$$g_n = D_{p,n} R^{-2n} \left\{ \sum_{\nu \in J} r_\nu s_\nu^{-p} e^{-a'(0)s_\nu/2} \left(\frac{R}{s_\nu} \right)^{2n} + o(1) \right\}. \quad (5.3)$$

If the numbers s_ν^2 , $\nu \in J$, are pairwise distinct, then

$$\limsup_{n \rightarrow \infty} \left(\frac{|g_n|}{n!} \right)^{1/n} = \frac{4}{R^2} > 0. \quad (5.4)$$

In particular $\widehat{G}$ cannot be the Taylor series of a holomorphic germ. Pairwise distinctness holds automatically if all the nearest poles lie strictly to the left of a_0 , that is, $\operatorname{Re}(a_0 - \omega_\nu) > 0$.

Proof. Subtract all principal parts at the nearest poles. The remainder is analytic on a disc of radius greater than R . For a principal part $r(a - \omega)^{-h}$, its contribution to (5.2) is

$$r \sum_{\ell \geq 0} \frac{D_{h,\ell} t^\ell}{(a(t) - \omega)^{2\ell+h}}.$$

Lemma 4.1 gives its coefficient asymptotic. If $h < p$, then

$$\frac{D_{h,n}}{D_{p,n}} \sim \frac{\Gamma(p)}{\Gamma(h)} (2n)^{h-p} \longrightarrow 0.$$

Lemma 4.3, with $R < R_1 < R_*$, makes the analytic remainder exponentially smaller than $D_{p,n} R^{-2n}$. This proves (5.3).

Set

$$A_\nu = r_\nu s_\nu^{-p} e^{-a'(0)s_\nu/2}, \quad \rho_\nu = (R/s_\nu)^2.$$

All A_ν are nonzero and all $|\rho_\nu| = 1$. Distinct squared distances mean distinct ρ_ν . The finite geometric-sum identity implies

$$\frac{1}{N} \sum_{n=0}^{N-1} \left| \sum_{\nu \in J} A_\nu \rho_\nu^n \right|^2 \longrightarrow \sum_{\nu \in J} |A_\nu|^2 > 0. \quad (5.5)$$

Thus the expression in braces in (5.3) is bounded and bounded away from zero along an infinite subsequence. Finally,

$$\left(\frac{D_{p,n}}{n!} \right)^{1/n} \longrightarrow 4$$

by Stirling's formula, or by the central-binomial coefficient estimate and the finite product relating $D_{p,n}$ to $D_{1,n}$. Equation (5.4) follows.

If $s_\nu^2 = s_\mu^2$ with $\nu \neq \mu$, then $s_\nu = -s_\mu$. This is impossible when both have strictly positive real part. A holomorphic germ has exponentially bounded Taylor coefficients and hence zero value of the limsup in (5.4), proving the final assertion. $\square$

5.1 From a formal heat expression to actual boundary derivatives

Suppose f_j is periodic and $a(t)$ is analytic with $\operatorname{Re} a(0) > 0$. For $\operatorname{Re} t \leq 0$ and sufficiently small t , define

$$G(t) = \sum_{j \geq 1} f_j e^{-ja(t)+j^2 t}, \quad \phi(a) = \sum_{j \geq 1} f_j e^{-ja}. \quad (5.6)$$

Every fixed derivative is uniformly dominated by a polynomial in j times e^{-cj} for some $c > 0$. Differentiation to the boundary is therefore legitimate. Expanding a fixed derivative, or collecting the finite Taylor coefficient of each exponential, shows that the Taylor jet of G is exactly (5.2). In particular, if G had a holomorphic continuation through zero, that continuation would have those coefficients. The same argument applies to exponentially bounded f_j whenever $\operatorname{Re} a(0)$ is sufficiently large.

This observation does not replace the divergent series by a convergent one. It only identifies each finite derivative, which is all the contradiction requires.

6 Rational phases force a natural boundary

6.1 The finite Fourier data

Fix a nonzero rational imaginary point

$$u_0 = \frac{2\pi ir}{m}, \quad r \in \mathbb{Z}, \quad m \in \mathbb{N}_{>0}, \quad |u_0| < r_0. \quad (6.1)$$

The sequence $f_j = e^{u_0 j^2}$ is m -periodic, since $(j+m)^2 - j^2 = 2jm + m^2$. Its generating function is rational in $z = e^{-a}$:

$$\phi(a) = \sum_{j \geq 1} f_j e^{-ja} = \frac{\sum_{j=1}^m f_j e^{-ja}}{1 - e^{-ma}}. \quad (6.2)$$

Equivalently, with the finite Fourier coefficients

$$\rho_k = \frac{1}{m} \sum_{j=1}^m \exp\left(\frac{2\pi i(rj^2 - kj)}{m}\right), \quad 0 \leq k < m, \quad (6.3)$$

we have

$$\phi(a) = \sum_{k=0}^{m-1} \frac{\rho_k}{e^{a-2\pi ik/m} - 1}. \quad (6.4)$$

Its poles are simple, lie on the imaginary axis, and have residues ρ_k at $2\pi ik/m + 2\pi i\mathbb{Z}$ whenever $\rho_k \neq 0$. At least one such residue is nonzero, because the finite Fourier transform is invertible and f_j is not the zero sequence. No nonvanishing theorem for a particular quadratic Gauss sum is needed.

6.2 Why an analytic implicit inverse is impossible

Suppose, for contradiction, that Q extends holomorphically across u_0 . Since $Q(u_0) \neq 0$ and $|Q(u_0)| < 1$, a local analytic logarithm gives

$$a(t) = -\log Q(u_0 + t), \quad \operatorname{Re} a(0) > 0. \quad (6.5)$$

On the left side of $t = 0$, the defining equation becomes

$$\sum_{j \geq 1} f_j e^{-ja(t)+j^2t} = u_0 + t. \quad (6.6)$$

The Taylor coefficients of the left side are those of the curved heat series generated by (6.2). All its poles lie strictly to the left of $a(0)$ in real part. Theorem 5.1 therefore gives a strictly positive factorial-normalized type. The right side has no coefficient of degree at least two. This contradiction proves that Q has no holomorphic continuation at u_0 .

Points (6.1) with $u_0 \neq 0$ are dense in the imaginary interval $|\operatorname{Im} u| < r_0$. A continuation across any point of that interval would give a continuation across nearby rational points. Hence every point, including zero, is a natural-boundary point. This proves Theorem 2.3(i).

Remark 6.1 (Why a fixed-amplitude argument is insufficient). For a fixed q , the series $\sum q^j e^{j^2 u}$ has a familiar lacunary structure after putting $z = e^u$. But the implicit identity uses $q = Q(u)$, not a constant q . The nonzero multiplier in (5.3) is what excludes cancellation by an analytic Q . This is the additional argument needed to pass from a partial theta function to an implicitly defined branch.

6.3 The natural boundary of the forward realization

By (3.4), Q maps D_- conformally onto $\Omega = Q(D_-)$ and maps its diameter homeomorphically onto Γ . Let $U = Q^{(-1)}$ on Ω . Suppose U had a holomorphic continuation across $q_0 = Q(it_0)$. Its derivative there would be the interior limit $1/Q'(it_0) \neq 0$. The holomorphic inverse function theorem would then produce a local holomorphic inverse of this continuation near it_0 . On the original side that inverse equals Q , contradicting the natural boundary just proved.

If Γ were a regular real-analytic arc near any of its points, an analytic parametrization with nonzero derivative would extend to a local holomorphic coordinate straightening the arc to a real interval. In that coordinate Q has real boundary values on a straight interval. Schwarz reflection extends it holomorphically across that interval; undoing the coordinate contradicts the same natural boundary. Thus Γ is nowhere real-analytic as a regular curve.

Its smooth jet at the origin nevertheless begins

$$Q(it) = it + 2t^2 - \frac{i}{2}t^3 - \frac{2}{3}t^4 + O(t^5). \quad (6.7)$$

A low-order polynomial plot can therefore look completely regular and almost parabolic. The obstruction is not visible in a finite jet.

7 An entire Borel transform and its fine sum

This section reproduces, in the quadratic specialization, the required summation facts. The signed Lagrange-block method comes from the repository's negative-ray article [2]; the simplified frequency bounds below make the present proof short.

7.1 Exact exponential blocks

Invert $q \mapsto K(q, u)$ while keeping u temporarily fixed and the target w independent. The coefficient form of Lagrange inversion [8] gives

$$q(w; u) = \sum_{k \geq 1} \frac{w^k}{k} [q^{k-1}] \left(\sum_{j \geq 1} q^{j-1} e^{j^2 u} \right)^{-k}.$$

Setting $w = u$ is formally legitimate. For $k \geq 2$ and an ordered composition $\mathbf{m} = (m_1, \dots, m_\ell)$ of $k - 1$ into positive parts, define

$$w_{k,\ell} = \frac{(-1)^\ell}{k} \binom{k + \ell - 1}{\ell}, \quad b_{k,\mathbf{m}} = \sum_{i=1}^{\ell} m_i^2 + k - 2. \quad (7.1)$$

The exact block representation is

$$\widehat{Q}(u) = u e^{-u} + \sum_{k \geq 2} u^k \sum_{\mathbf{m} \models k-1} w_{k,\ell(\mathbf{m})} e^{b_{k,\mathbf{m}} u}. \quad (7.2)$$

Indeed the exponent before simplification is $\sum_i (m_i + 1)^2 - (k + \ell)$, which equals the expression in (7.1). The important bounds are

$$2k - 3 \leq b_{k,\mathbf{m}} \leq k^2 - k - 1 < k^2, \quad k \geq 2. \quad (7.3)$$

Thus the only negative-frequency block is the initial $u e^{-u}$.

The uncanceled total mass of the k th block is

$$C_k = \frac{1}{k} \sum_{\ell=1}^{k-1} \binom{k + \ell - 1}{\ell} \binom{k - 2}{\ell - 1}, \quad C_1 = 1. \quad (7.4)$$

A second application of Lagrange inversion, or substitution in the quadratic equation, yields

$$\sum_{k \geq 1} C_k t^k = \frac{1 + t - \sqrt{1 - 6t + t^2}}{4}. \quad (7.5)$$

The coefficients are nonnegative and the value at $t = 1/6$ is $1/4$. Consequently

$$C_k \leq \frac{6^k}{4}, \quad k \geq 1. \quad (7.6)$$

For $\operatorname{Re} u \leq 0$, the blocks with $k \geq 2$ have modulus at most $C_k |u|^k$. Hence (7.2) converges normally on a small left half-disc and equals the literal inverse there, by ordinary Lagrange inversion in q and the uniqueness from Section 3.

7.2 A normalized Bessel kernel

For $k \geq 1$ and real b put

$$B_{k,b}(\zeta) = \sum_{m \geq 0} \frac{b^m \zeta^{m+k}}{m!(m+k)!}. \quad (7.7)$$

This is the integrated Borel transform of $u^k e^{bu}$. The probability density

$$p_k(v) = \frac{\Gamma(k+1)}{\sqrt{\pi} \Gamma(k+1/2)} (1 - v^2)^{k-1/2}, \quad -1 < v < 1,$$

has even moments

$$\int_{-1}^1 v^{2m} p_k(v) dv = \frac{(2m)!k!}{4^m m!(m+k)!}.$$

Expanding the exponential and using those moments proves

$$B_{k,b}(\zeta) = \frac{\zeta^k}{k!} \int_{-1}^1 e^{2\sqrt{b}\zeta v} p_k(v) dv. \quad (7.8)$$

The integral is independent of the square-root choice. In particular, for $b \geq 0$,

$$|B_{k,b}(\zeta)| \leq \frac{|\zeta|^k}{k!} e^{2\sqrt{b}|\operatorname{Re} \sqrt{\zeta}|}, \quad |B_{k,b}(-s)| \leq \frac{s^k}{k!}. \quad (7.9)$$

For $b < 0$, the same integral gives the less sharp but sufficient bound $|B_{k,b}(\zeta)| \leq |\zeta|^k e^{2\sqrt{|b||\zeta|}}/k!$.

7.3 Entire continuation and tube bounds

Apply (7.7) termwise to (7.2). On $|\zeta| \leq R$, equations (7.3) and (7.6) bound the k th block by

$$\frac{6^k}{4} \frac{R^k}{k!} e^{2k\sqrt{R}}.$$

The sum converges for every R , so the resulting Borel function is entire. Its Taylor coefficients at zero equal those of $\mathcal{B}_0 \hat{Q}$, because each ordinary coefficient uses only finitely many outer indices k .

For $\zeta \in T_\delta$,

$$|\operatorname{Re} \sqrt{\zeta}|^2 = \frac{|\zeta| + \operatorname{Re} \zeta}{2} \leq \delta. \quad (7.10)$$

This follows from the 2-Lipschitz property of $\zeta \mapsto |\zeta| + \operatorname{Re} \zeta$, which vanishes on the negative ray. Therefore

$$|\mathcal{B}_0 \hat{Q}(\zeta)| \leq |\zeta| e^{2\sqrt{|\zeta|}} + \frac{1}{4} \exp(6e^{2\sqrt{\delta}}|\zeta|), \quad \zeta \in T_\delta. \quad (7.11)$$

The first term is also bounded by a constant times an exponential in $|\zeta|$. This proves fine summability on every fixed-width tube.

7.4 Identification of the fine sum

For each fixed k, b , integrating the entire series in (7.7) termwise against $r^{-1}e^{-s/r}$ is justified by the integrable absolute series. It gives

$$\frac{1}{r} \int_0^\infty e^{-s/r} B_{k,b}(-s) ds = (-r)^k e^{-br}. \quad (7.12)$$

To exchange the *infinite block sum* with the integral, use the stronger ray estimate in (7.9):

$$\sum_{k \geq 2} \sum_{m=k-1} |w_{k,\ell} B_{k,b_{k,m}}(-s)| \leq \sum_{k \geq 2} \frac{6^k s^k}{4k!} \leq \frac{1}{4} e^{6s}.$$

The initial block is bounded by $se^{2\sqrt{s}}$. Thus dominated convergence applies when $0 < r < 1/6$, after also restricting r to the small inverse domain. Equation (7.12) then returns exactly (7.2) at $u = -r$. This proves $\mathcal{S}_\pi \hat{Q}(-r) = Q(-r)$. The same identity holds on the common complex domain by holomorphy.

8 Why no angular neighborhood can work

Lemma 8.1 (Angular growth produces a larger input sector). *Let B be holomorphic near zero and on $|\arg \zeta - \pi| < \eta$, and suppose $|B(\zeta)| \leq Ce^{H|\zeta|}$ on a smaller closed angular sector still containing π . Its Laplace transforms along those rays glue to a holomorphic function in an input sector centered at π with opening strictly greater than π . On the negative input ray it is the integral (2.2).*

Proof. Take $0 < \eta_1 < \eta$ on which the exponential bound is uniform. For a ray of direction $\theta \in [\pi - \eta_1, \pi + \eta_1]$, the integral

$$\frac{1}{u} \int_0^{e^{i\theta}\infty} e^{-\zeta/u} B(\zeta) d\zeta$$

converges whenever $\operatorname{Re}(e^{i\theta}/u) > H$. On overlaps, contour deformation between two sufficiently close rays is justified by the exponential bound: the large joining arc has exponentially decreasing integrand, and the small arc tends to zero. The integrals therefore agree and glue. Choose $0 < \eta_2 < \eta_1$. Every direction $|\arg u - \pi| < \pi/2 + \eta_2$ makes an angle strictly less than $\pi/2$ with an allowed Laplace ray; after decreasing the input radius uniformly on a slightly smaller closed sector, the convergence condition holds. The resulting opening is $\pi + 2\eta_2 > \pi$. $\square$

Proof of Theorem 2.3(iii). Suppose an exponential bound existed for $\mathcal{B}_0\hat{Q}$ in an angular neighborhood of π . Lemma 8.1 would produce a holomorphic sum on a sector of opening greater than π centered at the negative axis. By Section 7, this function agrees with Q on a negative real interval. The identity theorem makes it an analytic continuation of Q on the connected overlap with the left half-disc. The larger sector contains sufficiently small nonzero points of both imaginary rays, with neighborhoods crossing the imaginary axis. This contradicts Section 6.

Thus no such exponential bound exists. Since $\mathcal{B}_0\hat{Q}$ is bounded on compact subsets of the plane, the unbounded supremum in (2.6) must be witnessed at unbounded radii. This proves both the growth assertion and failure of angular summability for $\hat{Q}$.

Angularly 1-summable tangent-to-identity series form a group under composition, and their summation commutes with inversion; this is the standard nonlinear inversion theorem, in small-variable notation [4, Theorem 17.3]. Equivalently, it follows by changing from $u = 0$ to $z = \infty$, applying the tangent-to-identity inversion theorem there, and using the algebra property for reciprocals. If $\hat{U}$ were angularly summable in direction π , its formal inverse $\hat{Q}$ would be so after shrinking the arc. This contradiction proves the assertion for $\hat{U}$. $\square$

Corollary 8.2 (Divergence despite an entire Borel transform). *The series $\hat{Q}$ has radius of convergence zero, although $\limsup(|[u^n]\hat{Q}|/n!)^{1/n} = 0$.*

Proof. Entirety of $\mathcal{B}_0\hat{Q}$ gives the displayed zero limsup. A convergent power series has a Borel transform of finite exponential type in every direction, so would contradict Theorem 2.3(iii). $\square$

Remark 8.3 (Finite-plane resurgence versus infinity). No finite Borel singularity can explain the obstruction in this example: there are none for $\mathcal{B}_0\hat{Q}$. In a definition of resurgence based only on endless analytic continuation, the Borel transform is automatically resurgent. Assertions that it is “not resurgent” would therefore be incorrect without an additional growth convention. The precise conclusion is failure of angular 1-summability at π .

9 Rational amplitudes and robustness

9.1 A general quadratic-growth fine-summation lemma

The Borel estimate extends beyond rational amplitudes. Rationality will be used only for the natural-boundary obstruction.

Lemma 9.1 (Exponentially bounded amplitudes and quadratic slopes). *Suppose $c_1 \neq 0$, $|c_j| \leq MD^j$, and the real slopes satisfy*

$$-L \leq \lambda_j \leq L(1 + j^2)$$

for some positive M, D, L . Then the small left-side inverse of (2.9) has an entire integrated Borel transform, is finely summable in direction π , and is the corresponding fine sum. It is holomorphic in a small left half-disc and smooth on its diameter.

Proof. Choose a q -disc on which the geometric majorants and all their polynomially weighted derivatives converge. For sufficiently small u in the closed left half-plane, solve the equation by isolating $c_1 e^{\lambda_1 u} q$ and applying a uniform contraction to the remaining terms. Because $\lambda_j \geq -L$, $|e^{\lambda_j u}| \leq e^{L|u|}$ there. The derivative at $q = 0$ tends to $c_1 \neq 0$. This proves existence, uniqueness, holomorphy and, by the derivative argument of Section 3, smoothness.

The Lagrange blocks are indexed as before. Their scalar weights and frequencies are

$$w_{k,\mathbf{m}} = \frac{(-1)^\ell}{k} \binom{k+\ell-1}{\ell} c_1^{-k-\ell} \prod_{i=1}^{\ell} c_{m_i+1}, \quad (9.1)$$

$$b_{k,\mathbf{m}} = \sum_{i=1}^{\ell} \lambda_{m_i+1} - (k+\ell)\lambda_1. \quad (9.2)$$

The initial block is $c_1^{-1} u e^{-\lambda_1 u}$. The exponential coefficient bound, the number of compositions, and the binomial bound imply that the uncanceled total absolute weight of block k is at most C_0^k for a constant C_0 . The slope assumptions give, after enlarging C_1 ,

$$-C_1 k \leq b_{k,\mathbf{m}} \leq C_1 k^2. \quad (9.3)$$

For the upper bound use $\sum_i (m_i + 1)^2 \leq (\sum_i (m_i + 1))^2 \leq 4k^2$; for the lower bound use $\ell \leq k$ and the uniform lower bound on the slopes.

On any compact Borel disc the bounds from (7.8) are $|\zeta|^k \exp(O(k))/k!$, so the weighted Borel series converges normally and is entire. For nonnegative frequencies in T_δ , its terms are bounded by

$$C_0^k \frac{|\zeta|^k}{k!} \exp(2\sqrt{C_1 \delta} k).$$

For negative frequencies, use

$$2\sqrt{C_1 k |\zeta|} \leq k + C_1 |\zeta|$$

to obtain a bound $C_0^k |\zeta|^k e^{k+C_1 |\zeta|}/k!$. Summing over k yields an exponential bound on each fixed tube.

On the negative ray, the same estimates dominate the complete Borel sum by Ce^{Hs} for some C, H . Equation (7.12) and dominated convergence identify its Laplace transform with the convergent exponential-block inverse for small negative inputs. Its equality with the literal branch follows from the parameterized Lagrange formula and uniqueness. This proves the lemma without assuming positivity of the amplitudes. $\square$

9.2 Periodic twisting preserves rationality and infinite support

Let $\mathcal{A}$ be as in (2.7), and let f_j be a nonzero periodic sequence whose values are all nonzero. Then

$$\mathcal{A}_f(z) = \sum_{j \geq 1} c_j f_j z^j \quad (9.4)$$

is rational: expand f_j in its finite Fourier series and write $\mathcal{A}_f$ as a finite linear combination of $\mathcal{A}(\omega z)$, where ω ranges over roots of unity. If $\mathcal{A}_f$ were a polynomial, its coefficients would vanish for all large j . Since $f_j \neq 0$, the same would hold for c_j , so $\mathcal{A}$ would be a polynomial. Thus a nonpolynomial rational amplitude remains nonpolynomial after quadratic rational-phase twisting.

All finite poles of $\mathcal{A}_f$ lie at rotated poles of $\mathcal{A}$; cancellations can remove poles but cannot introduce smaller-modulus ones. If R_A is the radius of convergence of $\mathcal{A}$, then for $|q_0| < R_A$, every pole of $\phi(a) = \mathcal{A}_f(e^{-a})$ lies strictly to the left of $a_0 = -\log q_0$ in real part. Indeed a pole z_* gives $\operatorname{Re} \omega = -\log |z_*|$, whereas $\operatorname{Re} a_0 = -\log |q_0| > -\log R_A$. The poles form a finite union of vertical lattices and have finite orders. Theorem 5.1 applies, including possible higher-order poles.

9.3 Eventually quadratic slopes

Assume now (2.8) and suppose $\mathcal{A}$ is nonpolynomial. The assumptions of Lemma 9.1 hold, since the coefficients of any rational germ are exponentially bounded and the slopes are bounded below and grow at most quadratically.

Take a nonzero imaginary point u_0 sufficiently close to zero with

$$\alpha u_0 = \frac{2\pi i r}{m}. \quad (9.5)$$

Suppose that Q extends analytically across u_0 . Set $u = u_0 + t/\alpha$. Apart from finitely many terms, the left side of (2.9) is

$$e^{\gamma u} \sum_{j \geq 1} c_j e^{\alpha u_0 j^2} \left(Q(u) e^{\beta u} \right)^j e^{j^2 t}. \quad (9.6)$$

The discrepancy from the exceptional slopes is a finite analytic sum under the assumed analyticity of Q . The factor $Q(u_0)$ is nonzero, because the right side of (2.9) is $u_0 \neq 0$. Choose the half-disc small enough that $|Q(u_0) e^{\beta u_0}| < R_A$; this is possible because $Q(u) \rightarrow 0$ and β is real. An analytic logarithm of $Q(u) e^{\beta u}$ supplies an analytic curve $a(t)$ in the heat reduction.

The twisted amplitude is rational and nonpolynomial by the preceding subsection. Its nearest heat poles lie strictly to the left of $a(0)$. Theorem 5.1 shows that the heat expression in (9.6) cannot be analytic in t . But the defining equation, after subtracting the finite exceptional terms and dividing by $e^{\gamma u}$, says that it is analytic. This contradiction proves noncontinuity at every point (9.5). Density gives the entire small imaginary interval as a natural boundary.

Completion of Theorem 2.4. The preceding arguments prove the existence, Borel and natural-boundary assertions in the nonpolynomial case. Fine-sum identification and Lemma 8.1 exclude angular summability just as in the quadratic unit-amplitude model. Positive Taylor radius would imply angular summability and is therefore excluded as well.

If $\mathcal{A}$ is a polynomial, the kernel in (2.9) is a finite sum of entire functions of u and polynomials in q . It is holomorphic near $(0, 0)$, with q -derivative $c_1 \neq 0$. The holomorphic implicit function theorem gives a convergent inverse germ. Its Borel transform has finite exponential type in every direction and its usual sum agrees with the fine sum. All four stated conditions then hold. $\square$

10 Finite-action approximation does not detect the boundary

The natural boundary coexists with exceptionally strong approximation by analytic finite-action models. This is relevant to symbolic algorithms and to interpreting finite numerical evidence.

For the unit quadratic model put

$$K_J(q, u) = \sum_{j=1}^J q^j e^{j^2 u}, \quad K_J(Q_J(u), u) = u.$$

The same contraction used for Q defines Q_J on the same closed half-disc. Since the finite kernel is holomorphic across its diameter and its q -derivative remains nonzero there, each Q_J extends holomorphically across every point of that diameter. The size of the extension neighborhoods is not uniform in J .

Theorem 10.1 (Exponential approximation and exact finite jets). *For every $J \geq 1$,*

$$\sup_{\substack{|u| \leq 1/32 \\ \operatorname{Re} u \leq 0}} |Q(u) - Q_J(u)| \leq \frac{7680}{5983} 16^{-(J+1)}. \quad (10.1)$$

For each fixed integer $m \geq 0$ there is a constant C_m , independent of J , such that

$$\max_{0 \leq h \leq m} \sup_{\substack{|u| \leq 1/32 \\ \operatorname{Re} u \leq 0}} |Q^{(h)}(u) - Q_J^{(h)}(u)| \leq C_m (J+1)^{2m} 16^{-(J+1)}. \quad (10.2)$$

Furthermore,

$$[u^n] \widehat{Q}_J = [u^n] \widehat{Q}, \quad n \leq J. \quad (10.3)$$

Proof. The difference between the full and truncated contraction maps is at most

$$\frac{32}{31} \sum_{j>J} (1/16)^j = \frac{32}{31} \frac{16^{-(J+1)}}{1 - 1/16}.$$

Divide by $1 - 32/225$, the fixed-point stability denominator from (3.2). The resulting constant is $7680/5983$, proving (10.1).

For derivatives, every mixed kernel derivative of total order at most m introduces at most a constant times j^{2m} into the tail, up to a fixed factor from q -differentiation. Consequently its tail is bounded by $C'_m (J+1)^{2m} 16^{-(J+1)}$. To compare evaluation at Q and Q_J , use the mean-value estimate in the q variable; all needed kernel q -derivatives are uniformly bounded on a disc containing both values, because both actually lie in $|q| \leq 17/465 < 1/16$. Repeated implicit differentiation expresses the h th derivative of Q as a rational polynomial in mixed kernel derivatives of order at most h , the lower derivatives of Q , and $1/K_q$. The denominators stay uniformly away from zero for all J . Induction on h , using (10.1) at the initial step, proves (10.2). Additional q derivatives used only to compare evaluation points are uniformly bounded and multiply the already controlled difference $Q - Q_J$; they do not increase the power of J in the tail bound.

Finally, the omitted kernel terms have q -order at least $J+1$. Since the inverse starts with u , they contribute only from u -order $J+1$ onward in the formal implicit equation. Recursive uniqueness of the coefficients proves (10.3). $\square$

This theorem rules out two tempting but invalid inference patterns. Convergence of analytic approximants in every fixed differentiability norm on a one-sided domain does not supply analytic continuation of the limit. Likewise, agreement of arbitrarily long finite Taylor prefixes with convergent finite-action models does not supply a common convergence radius or angular Borel bound.

11 Examples, checks, and limitations

11.1 A nonlinear analytic curve and one dominant pole

Take

$$a(t) = \log 2 + \frac{3}{10}t + \frac{1}{7}t^2, \quad G(t) = \sum_{j \geq 1} 2^{-j} \exp\left(tj^2 - \frac{3}{10}jt - \frac{1}{7}jt^2\right).$$

The initial function is $\phi(a) = 1/(e^a - 1)$. The nearest pole to $a_0 = \log 2$ is the simple pole at zero with residue one. Theorem 5.1 gives

$$\frac{g_n(\log 2)^{2n+1}}{D_{1,n}} \longrightarrow 2^{-3/20}. \quad (11.1)$$

All g_n are rational: the moments $\sum_{j \geq 1} j^h 2^{-j}$ are integers for $h \geq 1$, and the curve coefficients are rational. The accompanying program computes g_n exactly before evaluating the normalization at 120-digit precision.

n	$g_n(\log 2)^{2n+1}/D_{1,n}$	ratio to $2^{-3/20}$
10	0.900426863367	0.999086159422
20	0.900850024301	0.999555685875
40	0.901052952323	0.999780848615
80	0.901152369379	0.999891158745

These values test the nontrivial multiplier, including a genuinely nonlinear curve. The quadratic coefficient of the curve changes finite orders but not the limit. The computation is not used in the proof.

11.2 Two equally near poles: do not divide by an oscillating leading term

Replace the amplitude 2^{-j} by $(-1/2)^j$, keeping the same curve increment. Now $\phi(a) = -1/(e^a + 1)$ and the nearest poles to $\log 2$ are $\pm i\pi$, both with residue one. Let

$$R = \sqrt{(\log 2)^2 + \pi^2}, \quad s_{\pm} = \log 2 \pm i\pi.$$

The theorem gives the additive asymptotic

$$\frac{g_n R^{2n}}{D_{1,n}} - \sum_{\sigma \in \{+, -\}} \frac{e^{-3s_{\sigma}/20}}{s_{\sigma}} (R/s_{\sigma})^{2n} \longrightarrow 0. \quad (11.2)$$

The leading sum oscillates and can be small. A relative error would be an unjustified strengthening.

n	normalized coefficient	nearest-pole model	difference
10	-0.4906207135	-0.454908339	-0.035712374
20	0.4717175341	0.4692056033	0.0025119308
40	-0.5598737788	-0.553093519	-0.0067802598
80	0.03428136919	0.03726718722	-0.002985818

The nonmonotone displayed differences are compatible with an additive $o(1)$ statement. The proof uses the positive Cesàro mean square (5.5), not a pointwise lower bound at every index.

11.3 An actual rational boundary point

The point $u_0 = 2\pi i/257$ lies in the prescribed half-disc diameter. With $J = 96$, fixed-point evaluation gives

$$Q_J(u_0) \approx 0.00119519260283291956428615710818 + 0.0244408539831444770352844651668i.$$

The analytic action-truncation bound (10.1) is less than 2.04×10^{-117} . This bounds the difference of the exact full and exact finite-action functions; it does *not* by itself include floating-point rounding error. The recorded 120-digit arithmetic is therefore labeled a diagnostic, not an outward-rounded enclosure. The theorem proves noncontinuability at this point from finite Fourier pole data, without using the numerical value of $Q(u_0)$.

11.4 Rational amplitudes covered by the dichotomy

The amplitudes $c_j = 1$, $c_j = j$, and $c_j = j^d$ for a fixed nonnegative integer d all have rational generating functions with infinite support. So do nonzero periodic amplitudes after any finite initial correction that preserves $c_1 \neq 0$. Each is covered by Theorem 2.4. Higher-order poles in the polynomial amplitude examples are the reason the heat-rigidity theorem was proved for arbitrary pole order rather than only simple poles.

Geometric amplitudes $c_j = \rho^j$, $\rho \neq 0$, merely rescale the q variable and also lie in the class. Finite changes of coefficients or slopes cannot eliminate the obstruction as long as the amplitude generating function remains nonpolynomial and the linear coefficient remains nonzero.

11.5 Why the rationality restriction cannot be dropped

Consider $c_j = e^{-j^3}$ and $\lambda_j = j^2$. On every compact set in (q, u) ,

$$|c_j q^j e^{j^2 u}| \leq \exp(-j^3 + Cj^2 + C'j),$$

so the kernel converges normally on all of $\mathbb{C}^2$. Its derivative with respect to q at $(0, 0)$ is $e^{-1} \neq 0$, and its inverse is an ordinary holomorphic germ at zero. This infinite-support example is outside the rational-amplitude class. It shows that the general slogan “infinitely many quadratic actions force a natural boundary” is false. The amplitude hypothesis must remain in the theorem.

11.6 What the exact verification establishes

The standard-library part of `verify.py` computes the formal fixed point and its inverse independently of the signed-block formula. It checks both compositions, the literal implicit coefficient residual, the block expansion, the frequency bounds, the first Schröder masses, finite-action prefix agreement, and the rational contraction constants. The recorded run checks all these identities through degree sixteen. No finite verification is represented as a proof of the infinite natural-boundary or summability statements.

12 Further research questions

12.1 Individual Borel rays and their exponential types

The negative open-arc question is now settled, but the set

$$E_Q = \left\{ \theta : \limsup_{r \rightarrow \infty} \frac{1}{r} \log^+ |\mathcal{B}_0 \widehat{Q}(re^{i\theta})| < \infty \right\}$$

remains to be determined. Theorem 2.3 excludes a uniform angular neighborhood at π ; it does not say that every neighboring individual ray fails. A useful next result would distinguish isolated finite-type rays from families with locally bounded type.

12.2 The largest shrinking neighborhoods of the negative ray

Fixed-width neighborhoods admit exponential bounds, whereas fixed-angle wedges do not. Determine a sharp width function $w(r)$ for domains $|\operatorname{Im} \zeta| \leq w(-\operatorname{Re} \zeta)$ far along the negative axis. The Bessel estimate and $|\operatorname{Re} \sqrt{\zeta}| \asymp |\operatorname{Im} \zeta|/\sqrt{|\operatorname{Re} \zeta|}$ suggest that parabolic and subparabolic scales are natural, but the uncanceled bound is not an optimality proof. This is a quantitative question about cancellation at Borel infinity.

12.3 The actual boundary jets of the nonlinear inverse

The rigidity theorem tests a hypothetical analytic amplitude; it does not compute the large derivatives of the actual nonanalytic Q at a rational boundary point. Determine their exact Gevrey type, dominant phases and possible transseries. A successful calculation must solve a nonlinear derivative recurrence rather than insert the nonexistent analytic Taylor germ of Q into (5.3).

12.4 Irrational boundary points and arithmetic dependence

Density proves noncontinuability at irrational points without analyzing their local coefficient growth. Do Diophantine approximation properties of $\operatorname{Im} u/(2\pi)$ affect the optimal derivative bounds, quasi-analytic classes, or local summation procedures there? Rational approximants give increasing periods, so uniform control in the period is the first technical target.

12.5 Where the finite-action singularities accumulate

Every Q_J extends across the diameter, yet the limit does not. Locate the nearest branch points or other inverse singularities of Q_J , determine their distance to the imaginary axis as $J \rightarrow \infty$, and test whether their distribution converges to a measure on the limiting boundary. This would explain how a natural boundary emerges from finite analytic models rather than merely proving that it does.

12.6 Beyond rational amplitudes

Classify amplitude generating functions for which analytic-curve rigidity still forces a natural boundary. Algebraic functions with nonintegral local exponents are a natural next class: a corresponding transfer lemma should replace the integer pole order by a complex local exponent. Entire amplitudes require a separate growth analysis, as the e^{-j^3} example demonstrates. The desired criterion must combine singularities of the amplitude generating function with the quadratic slope scale.

12.7 Higher-degree and nonpolynomial slopes

For $\lambda_j \sim j^d$, $d > 2$, the local formal operator is a higher-order heat operator and the natural Gevrey scale changes. Analytic-curve multipliers and cancellation among equal d th powers of pole distances require new analysis. The simple right-half-plane argument excluding opposite squared distances is special to $d = 2$. For noninteger powers, rational boundary phases no longer become periodic by the argument used here.

12.8 Acceleration compatible with the literal branch

The failure of an angular 1-sum need not prohibit acceleration or a summation scheme adapted to shrinking domains. Construct such a procedure from the formal coefficients, prove that it selects the literal left-side inverse, and state exactly what continuation or uniqueness property it supplies. An acceleration theorem should not silently claim analytic continuation through the proved natural boundary.

12.9 Formalization and finite certificates

A useful formalization starts with the finite coefficient identities, then proves the uniform domination in (4.5), the pole decomposition, and the Cesàro noncancellation lemma. These form a reusable analytic-curve rigidity module. A separate complex-analysis layer should establish smooth implicit branches, boundary noncontinuation, and Laplace-sector gluing. The existing repository infrastructure does not make those new analytic obligations automatic.

13 Conclusion

The negative-ray summation problem and the angular summation problem have different answers for quadratic exponential feedback. The inverse has an entire Borel transform, and its Laplace integral is controlled in fixed-width Borel neighborhoods. Nevertheless a smooth natural boundary prevents every angular enlargement around the negative direction. The obstruction survives finite coefficient and slope perturbations and, more generally, gives a finite-support versus natural-boundary dichotomy for rational amplitudes with eventually quadratic slopes.

The mechanism is not merely that a partial theta kernel is singular. Its singularity cannot be removed by making its amplitude an analytic unknown: each nearest heat pole retains a nonzero analytic-curve multiplier, and the finite pole contributions cannot cancel permanently. This closes the specific implicit-cancellation gap required to answer the repository's open-arc question.

A Proof-dependency and claim audit

The natural-boundary theorem for the unit model depends only on the uniform implicit construction, the factorial composition lemma, analytic remainder majorization, the finite-pole Cesàro argument, and elementary periodic Fourier inversion. It does not use the earlier regularity article's large-order asymptotics, a general transseries realization theorem, or a numerical computation.

The entire-Borel and fine-summation statements are proved directly from Lagrange inversion and the normalized Bessel integral. The block representation is credited to the earlier repository work in its general form. The angular obstruction for $\hat{Q}$ then uses only elementary Laplace contour deformation and the identity theorem. The extension of that obstruction to the formal forward series $\hat{U}$ uses the standard angular summability inversion theorem [4]. The broader rational-amplitude classification uses the same elementary proof chain with a weighted block estimate and rational periodic twisting.

The nowhere-real-analytic assertion concerns the boundary *curve*, not just one parametrization, and uses local analytic straightening and Schwarz reflection. The finite-action theorem proves convergence on a closed one-sided domain; it does not supply a common two-sided holomorphic neighborhood. The numerical tables are diagnostics, not interval-certified real inequalities or substitutes for the proofs.

No Lean build was attempted and no Lean proof file is included. No repository branch or file was modified. The inspected research packages are themselves marked unmerged and not formalized in the repository README [1]. The present article should receive an independent mathematical review before being treated as an audited result.

B Reproducibility and source provenance

The package contains the editable `article.tex`, the compiled `article.pdf`, `verify.py`, generated exact and numerical JSON results, the two small TeX table inputs, the captured verification output, a README, and a source-provenance note. The TeX does not depend on a local copy of ProveIt or on custom fonts.

The commands used are

```
python verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact checks use Python’s `fractions.Fraction`. The numerical diagnostics use `mpmath` 1.3.0 at 120 decimal digits. The program makes no network requests. Its order can be increased up to twenty with `-max-order`; the independent composition enumeration has exponential cost and is intentionally bounded.

The following repository files were directly read at the pinned commit:

- `Analysis/Transseries/docs/series-and-transseries/README.md`, including its descriptions of the recently filed feedback articles;
- `Negative_Ray_Summation_Exponential_Feedback/article.tex` under that directory, including its definitions, Lagrange blocks, Bessel representation and explicit angular-summability question.

The source review was focused, not an exhaustive audit of every statement in the repository’s large transseries volumes. The reference to the regularity article supplies context as recorded in the directly read index and negative-ray paper; the present proof does not depend on its sharp asymptotic theorem.

References

- [1] V. Reshetnikov, *ProveIt: series and transseries*, repository index, commit 0c973d8f5e7ef4550f4df1bf486d890037fd9f14, accessed 29 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/0c973d8f5e7ef4550f4df1bf486d890037fd9f14/Analysis/Transseries/docs/series-and-transseries/README.md>.
- [2] *Negative-Ray Summation of Countable Exponential-Feedback Transseries*, research draft prepared for V. Reshetnikov, 29 September 2026, ProveIt, same pinned commit, directory `Analysis/Transseries/docs/series-and-transseries/Negative_Ray_Summation_Exponential_Feedback/`, `article.tex`. In particular, the research-agenda subsection “Angular summability of the quadratic model”; the direct source range read for that question was lines 1330–1410.
- [3] *A Sharp Regularity Classification for Countable Exponential-Feedback Transseries*, research draft in ProveIt, directory `Exponential_Feedback_Regularity_Classification/` under

the same series-and-transseries group, as described in [1, 2]. Contextual reference; its sharp coefficient asymptotics are not a dependency here.

- [4] D. Sauzin, *Introduction to 1-summability and resurgence*, arXiv:1405.0356, 2014. Sections 7 and 9 distinguish fine summability from summability in an arc; Section 17 proves the nonlinear group and inverse properties. <https://arxiv.org/abs/1405.0356>.
- [5] D. A. Lutz, M. Miyake, and R. Schäfke, On the Borel summability of divergent solutions of the heat equation, *Nagoya Mathematical Journal* **154** (1999), 1–29. DOI: [10.1017/S0027763000025289](https://doi.org/10.1017/S0027763000025289).
- [6] S. Michalik and B. Podhajecka, *The Stokes phenomenon for certain partial differential equations with meromorphic initial data*, arXiv:1603.04209, 2016. <https://arxiv.org/abs/1603.04209>.
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- [8] I. M. Gessel, Lagrange inversion, *Journal of Combinatorial Theory, Series A* **144** (2016), 212–249; arXiv:1609.05988. <https://arxiv.org/abs/1609.05988>.